HAMZA EL MAHJOUR (PH.D.)

Interested in Google Visiting Researcher Program

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Tangier, MOROCCO Check my Google Scholar Profile

EXPERIENCE

Professor of Mathematics & Researcher

FPL-Abdelmalek Essaadi University

Nov 2020 – Ongoing Larache,Morocco

- Teaching undergraduate students : calculus, linear algebra, differential equations.
- Teaching also Msc. students : Stochastic Calculus, numerical simulations of ODEs and SDEs

Primary/Middle school mathematics' Teacher

Jile Biladi - Savoir

Month 2013 – 2019 Tangier,Morocco

- I provided full courses from 4 graders to 9 graders.
- I supervised some new mathematics teachers to help them better integrate our teaching staff.

PROJECTS

Regularized reconstruction of dynamic PET images

U930/Inserm

6 months

- Improvement of existing OSEM code.
- Encouraging results when the processed images are compared to the ground truth images.

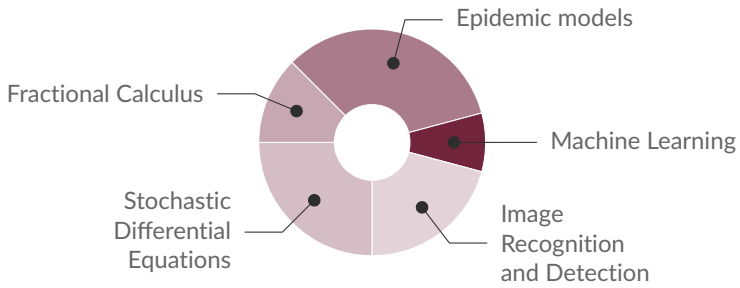
Epidemic modelling with memory effects

Applied Mathematics Laboratory - FST - UAE

3 years

- Thanks to fractional calculus, we can consider that the solutions are not memory less.
- Many inexact results on FDEs are revisited using rigorous mathematical proofs.
- The first attempt of real applications was on COVID-19 data sets in Morocco during the beginning of the pandemic.

RESEARCH INTERESTS



MY LIFE PHILOSOPHY

"Make your current day better than the previous one."

HOBBIES

Reading
Fiction and biographies

Sports
Tennis and chess

SKILLS

Adaptive | Motivator | Open-minded

C++ | Matlab | Statistical Analysis | Machine Learning

LANGUAGES

Arabic	●●●●●
French	●●●●●
English	●●●●●
Spanish	●●●○○

EDUCATION

Ph.D. in Probability and Statistics

FSTT-Abdelmalek Essaadi University

Sept 2014 – Nov 2019

Thesis title: Modélisation stochastique appliquée a l'épidémiologie des maladies infectieuses.

M.Sc. in Applied Mathematics

UFR Math-Université de Rennes 1

Sept 2011 – Sept 2013

B.Sc. in Pure Mathematics

UFR Math- Université de Rennes 1

Sept 2009 – June 2011

PUBLICATIONS

Journal Articles

- Lahrouz, A, H El Mahjour, A Settati, M Erriani, et al. (2022). “Bifurcation from an Epidemic Model in the Presence of Memory Effects”. In: *International Journal of Bifurcation and Chaos* 32.06, p. 2250077.
- Lahrouz, A, A Settati, et al. (2022). “A stochastic Sir epidemic evolution model with non-concave force of infection: Mathematical modeling and analysis”. In: *Stochastics and Dynamics*, p. 2250016.
- Assadouq, Abdelaziz, Hamza El Mahjour, and Adel Settati (2020). “Qualitative behavior of a SIRS epidemic model with vaccination on heterogeneous networks”. In: *Ital. J. Pure Appl. Math* 43, pp. 958–974.
- Lahrouz, Aadil, Adel Settati, Hamza El Mahjour, Mustapha El Jarroudi, et al. (2020). “Global dynamics of an epidemic model with incomplete recovery in a complex network”. In: *Journal of the Franklin Institute* 357.7, pp. 4414–4436.
- Lahrouz, A, H El Mahjour, A Settati, and A Bernoussi (2018). “Dynamics and optimal control of a non-linear epidemic model with relapse and cure”. In: *Physica A: Statistical Mechanics and its Applications* 496, pp. 299–317.

Conference Proceedings

- Lahrouz, Aadil, Adel Settati, Hamza El Mahjour, and Abdelhadi Akhrif (2014). “On the threshold of the SIS epidemic model with random perturbation”. In: *Nonlinear PDE & Applications : Theoritical and Numerical Study. CIMPA WORKSHOP*. Tangier,Morocco, pp. 186–189.

TEACHING

Vimeo
@ Non-profit channel
https://vimeo.com/user131894536

Official Website
@ Free downloadable ressources
https://hamzaelmahjour.github.io